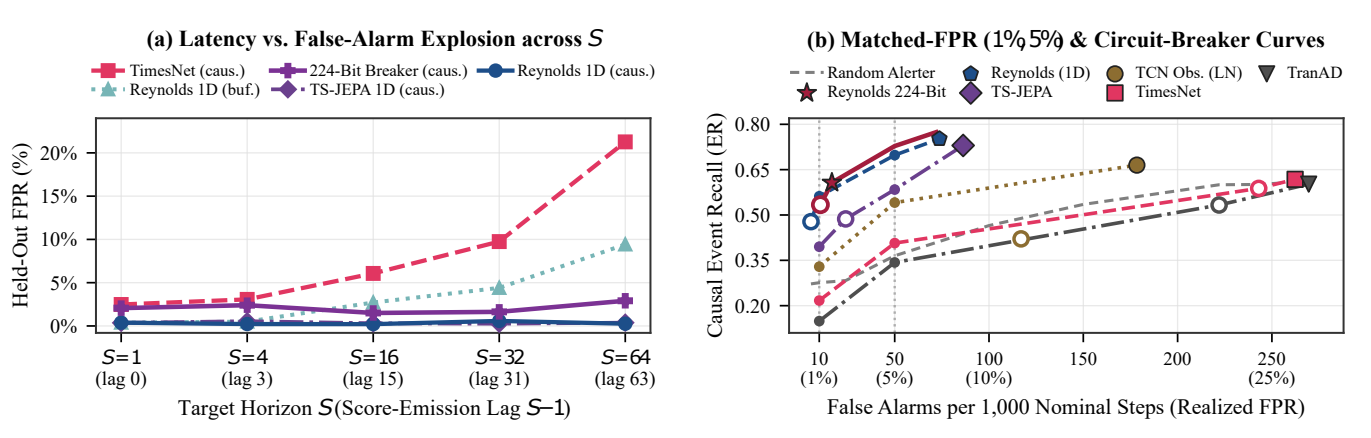

(a) Latency vs. False-Alarm Explosion across S
TimesNet (caus.)
224-Bit Breaker (caus.)
Reynolds 1D (caus.)
Reynolds 1D (buf.)
TS-JEPA 1D (caus.)
Held-Out FPR (%)
20%
15%
10%
5%
0%
S=1
(lag 0)
S=4
(lag 3)
S=16
(lag 15)
S=32
(lag 31)
S=64
(lag 63)
Target Horizon S (Score-Emission Lag S−1)
(b) Matched-FPR (1% 5%) & Circuit-Breaker Curves
Random Alerter
Reynolds 224-Bit
Reynolds (1D)
TS-JEPA
TCN Obs. (LN)
TimesNet
TranAD
Causal Event Recall (ER)
0.80
0.65
0.50
0.35
0.20
10
(1%)
50
(5%)
100
(10%)
150
200
250
(25%)
False Alarms per 1,000 Nominal Steps (Realized FPR)